

Adaptive Traffic Signal Control Using Deep Q-Networks: A Comparative Study with Webster's Fixed-Time Method

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Abstract—Traffic congestion at urban intersections remains a pressing challenge, causing significant delays and safety risks. While fixed-time signal controllers are widely used, many evaluations of adaptive traffic signal control (ATSC) use controllers with identical green light durations across all phases as the baseline. To ensure rigorous assessment, this study evaluates a Deep Q-Network (DQN)-based ATSC against a Webster-based fixed-time controller, a well-established traffic engineering benchmark. Using the Simulation of Urban MObility (SUMO), a DQN agent is trained with lane occupancy states to select optimal signal phases, aiming to minimize cumulative vehicle waiting times. The evaluation is conducted under varying demand scenarios of 750, 1,500, and 2,000 vehicles over a 1.5-hour period. Results show that the DQN-based ATSC consistently outperforms the Webster baseline at higher vehicle volumes, reducing total waiting times by more than 66%, while slightly better performance at lower volumes. These findings highlight the potential of reinforcement learning-based controllers to improve intersection efficiency and the importance of using established fixed-time methods as baselines for robust evaluations.

Index Terms—Adaptive traffic signal control, Deep Q-Network, Traffic congestion management, SUMO simulation, Webster's method

I. INTRODUCTION

Vehicle congestion in road traffic has raised safety risks, with studies showing crash occurrence likelihood in congested conditions is approximately six times higher than in free-flowing traffic. [14] Beyond safety, congestion also causes significant travel delays, which remain inadequately addressed in densely populated cities. Data from OICA [5] indicate that worldwide vehicle sales have increased by more than 19% since 2020. This growth may further intensify congestion if no effective solutions are implemented. Therefore, effective traffic management strategies are essential to address existing congestion and prevent its future escalation.

Most intersections are still managed using fixed-time controllers, where green durations remain constant regardless of traffic demand. These systems cannot adapt to fluctuating demands since they assume traffic demand remains similar at all times [3]. However, traffic systems are highly dynamic,

especially in urban areas, where disturbances such as construction sites or collisions can suddenly alter traffic volumes. In such cases, a system capable of adapting to real-time changes would be more effective [9].

Research on adaptive traffic signal control (ATSC) spans several categories of methods. Traditional fixed-time approaches, such as Webster's formula, [13] optimize cycle lengths and green splits based on average observed volumes but cannot respond to real-time fluctuations. Actuated and coordinated controllers extend these systems by incorporating sensor feedback and synchronization among adjacent intersections to improve local and corridor-level flow efficiency. More recent data-driven approaches, such as fuzzy logic control and model predictive control (MPC), further enhance adaptability but often require explicit traffic models and extensive calibration.

Reinforcement learning (RL) methods overcome these limitations by allowing the controller to learn optimal policies directly from interaction with the environment. Among RL algorithms, Deep Q-Networks (DQNs) [8]—which combine Q-learning [12] with deep neural networks—have become one of the most widely adopted for traffic signal control. DQN-based controllers can handle high-dimensional state spaces (e.g., vehicle positions or queue lengths) and learn adaptive signal strategies without pre-defined traffic models. Consequently, numerous studies have explored DQNs for adaptive signal optimization and as baselines for evaluating newer RL variants [7], [11], [15].

However, most existing evaluations compare RL controllers against simplistic fixed-time baselines that assign equal green durations to all phases, underestimating the effectiveness of optimized classical approaches. Recent works such as Lv et al. [7] compared DQN-based adaptive controllers with Webster-based fixed-time methods for signal timing optimization, while Wang et al. [11] conducted similar comparisons focusing on carbon dioxide emission reduction. These studies demonstrate growing interest in benchmarking RL-based controllers against classical traffic-engineering baselines. In this work we present rigorous evaluation of a standard DQN-based adaptive traffic signal controller against a Webster-based fixed-time controller, where green splits are proportionally allocated to observed demand. This comparison highlights both the strengths and

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limitations of the DQN-based ATSC relative to a well-established, demand-responsive benchmark.

II. METHODOLOGY

The methodology for this study is structured around the design, training, and evaluation of a DQN-based adaptive traffic signal control system. We describe the simulation environment and traffic flow generation process, followed by the formulation of the state representation, action space, and reward function. The architecture of the learning model and the training procedure are then detailed. Finally, the proposed controller is compared against a fixed-time baseline to assess its performance under varying traffic conditions. All the experiments including the training were conducted on a Windows 11 Pro system equipped with an Intel Core i7-12700 CPU (2.10 GHz), 32 GB RAM, and an NVIDIA GeForce RTX 3080 Ti GPU (12 GB VRAM).

The evaluation is done under the scenarios described in Table I. Depending on the traffic scenario, Webster's formula yields different green durations for each signal phase, and the fixed-time controller cycles through these phases sequentially. In contrast, the DQN-based controller uses the same green light duration for all phases within a given scenario, and this duration is taken from one of the green phase lengths calculated by Webster's method.

TABLE I
TRAFFIC SCENARIOS AND CONTROL SETTINGS. THE DQN AGENT USES THE SHOWN GREEN TIMES; THE FIXED-TIME BASELINE FOLLOWS WEBSTER'S METHOD.

Scenario	Vehicles (1.5 h)	DQN green light duration
8.3 veh/min	750	10 s
16.7 veh/min	1,500	10 s
22.2 veh/min	2,000	15 s

A. Simulation Environment

To train and evaluate the proposed adaptive traffic signal control system, the SUMO [2] traffic simulator is used. The simulation environment requires two main components: (i) a network file, which defines the geometry and structure of the road layout, and (ii) a route file, which specifies vehicle flows and movement patterns within the network.

1) *Network*: The network consists of a four-way intersection with traffic entering from the North, South, East, and West directions. Each direction consists of: (i) two through lanes dedicated to vehicles moving straight through, (ii) one shared through/right lane used by vehicles either going straight or turning right, and (iii) a dedicated left-turn lane reserved exclusively for vehicles making a left turn.

The intersection signal plan consists of green phases permitting through movements and right turns along the east-west and north-south approaches, complemented by exclusive left-turn phases in each direction. Every green phase is followed by its respective yellow interval to facilitate safe clearance, after which an all-red phase is employed as a clearance period. The phases are shown from Figures 2a - 2i.

The selected four-way intersection layout represents a common urban junction configuration with a mix of through, left-turn, and right-turn movements. This design was chosen because it captures the essential operational complexity of typical signalized intersection. While this setup provides a realistic baseline for comparison, results may vary under different geometries or network scales. For example, multi-intersection networks would introduce coordination challenges requiring additional state information and potentially multi-agent control strategies. Similarly, intersections with fewer turning lanes or simpler phase structures might reduce the relative advantage of adaptive controllers, since traffic dynamics would be less variable. Thus, the present configuration offers a balanced test scenario.

2) *Route*: Traffic routes are generated programmatically based on the network topology, where the junction with a traffic light. All possible routes from incoming to outgoing edges are enumerated, and each route is assigned a weight to control the likelihood of vehicle assignment. Straight routes (e.g., north to south) are assigned a weight of 6, while turning routes are assigned a weight of 2.

Vehicle departure times are distributed over the simulation period according to a Weibull distribution, following the approach described in [6], which better models realistic traffic flows compared to uniform spacing. All vehicles are defined as the same SUMO vehicle type, representing passenger cars with an acceleration of 2.6 m/s², a deceleration of 4.5 m/s², a driver imperfection factor of 0.5, a vehicle length of 5 m, a minimum gap of 2.5 m, and a maximum speed of 50 m/s. These vehicle parameters represent human-driven vehicles (HDVs), not autonomous vehicles. The driver imperfection factor in SUMO captures reaction variability and imperfect compliance with traffic signals, representing human-driven variability.. The generated trips and corresponding route definitions are then written to a SUMO-compatible XML route file for simulation.

B. State Representation

To enable the agent to perceive the traffic conditions at the intersection, we followed a state representation designed by [10], in which incoming lanes are discretized into a series of cells that record vehicle presence.

Each incoming approach (North, South, East, West) at the intersection is divided into multiple lanes, including three through/right lanes and one left-turn-only lane. The portion of the road considered for state representation extends 750 meters from the stop line of the intersection. This distance is discretized into 10 non-uniform cells per lane, with cell size increasing with distance from the stop line [10]. This non-uniform distribution balances detection accuracy near the intersection and computational efficiency further away.

Specifically, the cells closest to the intersection are shorter to accurately capture vehicles preparing to cross, while those further away are longer. Table II shows the boundaries of the cell intervals.

Each vehicle is mapped into a corresponding lane group and cell index, forming a unique position identifier in the

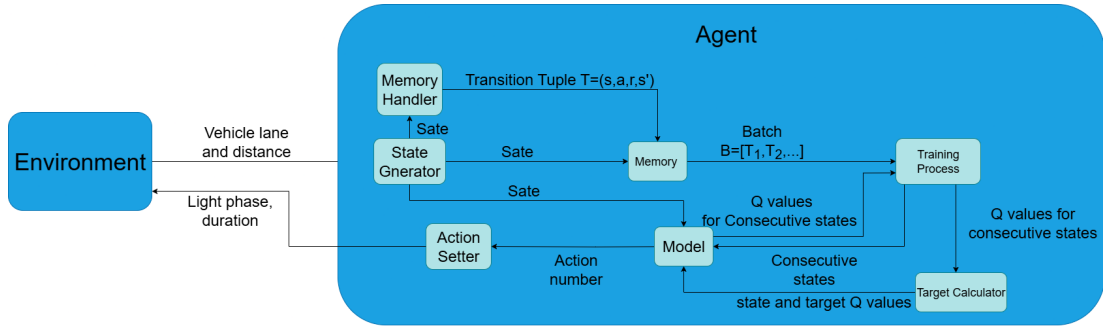


Fig. 1. Simulation and Training Workflow.

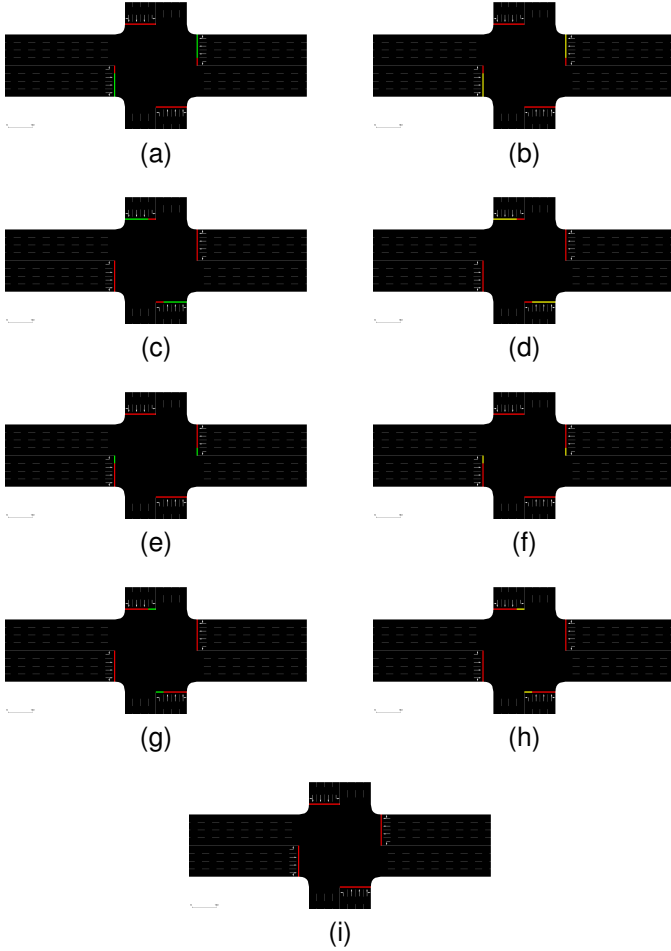


Fig. 2. (a) East West Green Phase, (b) East West Yellow Phase, (c) North South Green Phase, (d) North South Yellow Phase, (e) East West Left Green Phase, (f) East West Left Yellow Phase, (g) North South Left Green Phase, (h) North South Left Yellow Phase, (i) All Red (Clearance) Phase.

state vector. There are 8 lane groups representing the four directions (W, N, E, S), each with separate encoding for straight/right lanes and left-turn-only lanes. Each lane group contains 10 cells, resulting in a total of 80 binary features for the intersection state. A cell value of 1 indicates that a vehicle occupies the cell, while 0 indicates no presence.

TABLE II
CELL INDEX TO DISTANCE MAPPING.

Cell Index	Range (m)	Cell Index	Range (m)
0	0–7	5	40–60
1	7–14	6	60–100
2	14–21	7	100–160
3	21–28	8	160–400
4	28–40	9	400–750

The algorithm iterates over all detected vehicles using the SUMO and TraCI interface. For each vehicle, the following information is obtained:

Lane ID: To determine the lane group.

Lane position: Distance from the stop line, inverted and mapped to a cell index.

This binary occupancy state representation allows the DQN to process the spatial distribution of traffic approaching the intersection and make phase decisions accordingly.

C. Action Space

The action space represents the set of discrete decisions available to the agent. In this system, it consists of four predefined traffic signal phases, with each action corresponding to a specific direction or combination of directions receiving the green light [10]. Each action corresponds to a discrete decision on *which signal phase* to activate, not the duration of the green light. The duration of each green phase (T_g) is fixed within a scenario, as listed in Table I, while the agent's role is to adaptively select the next phase based on the current traffic state. Each action is represented by an integer in the range 0 to 3, where they are mapped to specific green phases:

- Action 0: East-West Green Phase
- Action 1: East-West Left Green Phase
- Action 2: North-South Green Phase
- Action 3: North-South Left Green Phase

The corresponding green phases are illustrated in Figure 3. These predefined, conflict-free phases eliminate the possibility of unsafe configurations, such as simultaneous green signals for conflicting directions. At any given time, only one phase is active, with all other approaches remaining red. Only one of these phases can be active at a time; simultaneous green signals for multiple conflicting directions are not permitted.

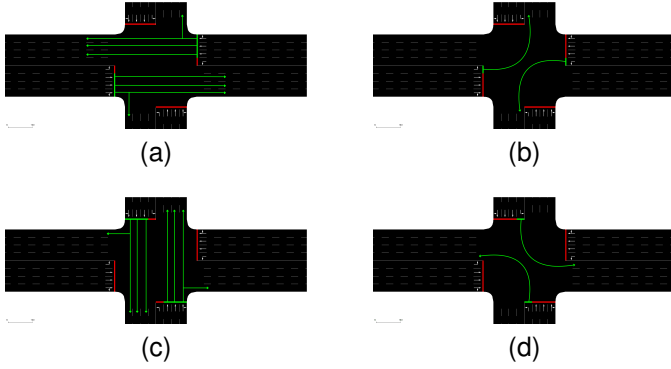


Fig. 3. (a) East West Green, (b) East West Left Green (c) North South Green, (d) North South Left Green.

This ensures that at any decision step, exactly one green phase is applied while all other approaches remain red. To ensure safe transitions between phases, each phase change follows a fixed inter-phase sequence, including both yellow and all-red intervals set to 3 seconds, in accordance with MUTCD (Manual on Uniform Traffic Control Devices) minimum requirements for yellow change intervals and established red clearance practices [1] [4].

Yellow phase — displayed for the lanes losing right-of-way, allowing vehicles to clear the intersection.

All-red phase — all approaches remain red for a short duration to ensure the intersection is fully cleared.

Next green phase — the new direction(s) receives right of way.

This sequence is applied consistently to all phase transitions.

D. Reward

The reward is based on the cumulative waiting time of all vehicles in the incoming lanes of the intersection [6] [10]. At each action step, the waiting times of all vehicles on the incoming roads are summed and stored. The reward R_t for action step t is then computed as:

$$R_t = W_{t-1} - W_t$$

where W_{t-1} is the cumulative waiting time at the previous step and W_t is the cumulative waiting time at the current step. A positive reward indicates a reduction in total waiting time, while a negative reward reflects an increase.

E. Model

In DQN-based Q-learning, the approximation of Q-values that represent the expected future rewards for possible actions is carried out by a deep neural network. In our implementation, inspired by [10], the network is a fully connected feed-forward architecture. The input layer corresponds to the state dimension, followed by four hidden layers of 400 neurons each with ReLU activation. The output layer produces Q-values for all possible actions with a linear activation. The model takes the generated state as input and outputs the Q-Values of all the possible actions.

F. Training Procedure

The training procedure follows the methodology described in [10], with the modification that training is conducted for 100 episodes, as the rewards were observed to stabilize beyond this point. Each episode simulated 1 hour and 30 minutes of traffic at SUMO's default resolution of 1 second per step, resulting in 5400 steps per episode. At each decision step, the agent observes the current state S_t , selects an action A_t according to an ϵ -greedy policy, executes the corresponding green phase, and receives an immediate reward R_t . The transition (S_t, A_t, R_t, S_{t+1}) is stored in a replay memory buffer.

Action selection follows an ϵ -greedy exploration strategy. The exploration rate ϵ is initialized to 1.0 and decayed linearly with each episode according to:

$$\epsilon = 1.0 - \frac{\text{episode}}{\text{total_episodes}} \quad (1)$$

ensuring a smooth transition from full exploration at the beginning of training to full exploitation by the final episode.

After the completion of each episode, mini-batches of size 100 are sampled from the replay buffer to update the Q-network parameters. The network is trained by minimizing the mean squared error between predicted and target Q-values, where the target is defined as:

$$Q_{\text{target}} = R_t + \gamma \max_{a'} Q(S_{t+1}, a'; \theta), \quad (2)$$

with γ denoting the discount factor and θ representing the parameters of the Q-network. The optimizer used is Adam with a fixed learning rate of 0.001, and training is carried out with 800 epochs after each episode. The discount factor of $\gamma = 0.75$ is used for our training process.

This process is repeated for 100 episodes with a duration of 5400 seconds each, corresponding to approximately 6.25 days of simulated traffic. To improve generalization, new traffic flows are generated for each episode, ensuring that the agent encountered diverse traffic scenarios during training.

G. Fixed-Time Signal Comparison

To evaluate the model's performance against a fixed-time traffic signal controller, a route file containing all traffic flow information is first generated. Based on this traffic flow data, the optimal cycle length and the green time allocation for each phase were calculated using Webster's formula [13]. In certain cases, the computed green duration for a phase is very short, sometimes less than five seconds. Such short intervals are impractical in real-world scenarios, as drivers and pedestrians may not have sufficient time to react to the signal change [1]. To address this issue, the calculated green durations are rounded up to the nearest higher multiple of five seconds (e.g., 3 seconds would be rounded to 5 seconds, while 7 seconds would be rounded to 10 seconds). This adjustment ensures that green intervals are long enough to be perceptible and usable, thereby matching driver behavior and expectations.

Using the same route file, two separate simulations are conducted: one with the trained model and one with the fixed-time controller.

For the model-based simulation, the process mirrors the training procedure:

- 1) The current state S_t is provided to the model.
- 2) The model outputs an action A_t , representing the selected green phase.
- 3) The simulation runs for the duration of that green phase (T_g seconds) and the queue length is saved for every simulation step. In our case it is one second.
- 4) The new state S_{t+1} is observed, and the average wait and reward are recorded.
- 5) The process repeats with S_{t+1} as the input to the model.

For the fixed-time simulation, the state S_t is still collected, and metrics are recorded at each decision step. However, instead of using the model to select A_t , the controller follows a predefined cyclic order of phases. The actions are mapped to integers between 0 and 3, and the next action is determined by:

$$A_{t+1} = (A_t + 1) \bmod 4$$

This ensures the next consecutive phase in the fixed-time cycle is always selected.

III. RESULTS AND ANALYSIS

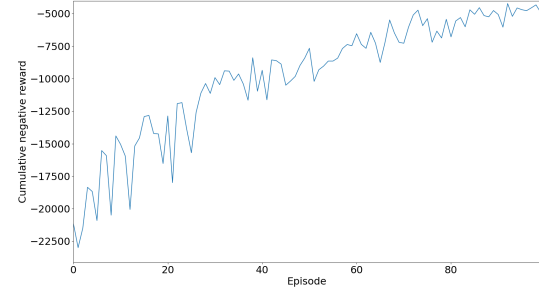
The performance of the DQN-based ATSC is evaluated against a Webster's method fixed-time controller under identical traffic demand scenarios. Both controllers operated on the same vehicle flows and arrival distributions, ensuring that observed differences were attributable solely to the control strategy. In high traffic scenario a green light duration of 15 seconds is used to allow more vehicles to pass through the intersection.

A. Cumulative Negative Reward Progression

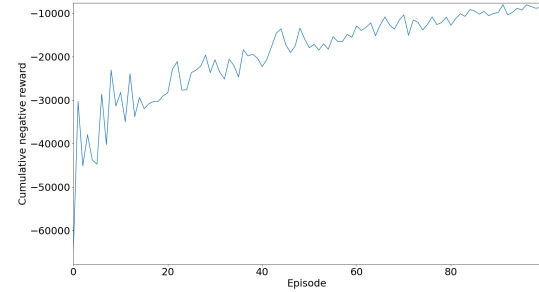
During training, cumulative negative rewards were monitored to assess policy improvement. At each action step, only negative rewards are accumulated, and the totals are aggregated across an episode. Figures 4a–4c illustrate the model's learning progress across traffic scenarios, where a reduction in cumulative penalties reflects improved policies.

B. Queue Length

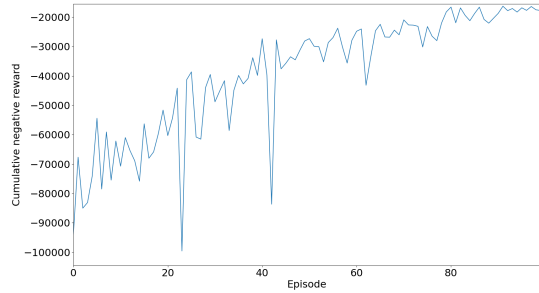
Queue lengths are recorded at one-second resolution for both controllers across all traffic scenarios. Figures 5a - 5c show the queue lengths of the compared controllers for one of the 10 simulations that was run. Observation of the plots show that the DQN-based ATSC consistently maintains shorter queues under low- and high-demand conditions. In the 8.3 veh/min scenario, both controllers produce comparable results, whereas in the 22.22 veh/min scenario, the DQN-based approach outperforms the fixed-time controller. We can see the gradual increase in the difference in performance of the two approaches as the number of vehicles in the simulation increases.



(a)



(b)



(c)

Fig. 4. Cumulative Negative Reward vs. Episode (a) 8.3 veh/min traffic scenario, (b) 16.7 veh/min traffic scenario, (c) 22.2 veh/min traffic scenario.

C. Overall Total Wait Time

The total accumulated wait time for all vehicles is measured across scenarios. Total wait times were collected for 10 different simulations. We collected total wait times for 10 independent simulations (different random seeds) and performed a paired two-tailed t-test on the sample of 10 paired observations. The results were shown in Table III. The DQN controller consistently outperforms the fixed-time approach, with improvements increasing under heavier congestion. At low traffic (8.3 veh/min), DQN achieves a modest 2.5% reduction in waiting time (12,670s vs. 12,999s, $p = 0.041$). Under moderate traffic (16.7 veh/min), the improvement increases to 23.4% (29,361s vs. 38,310s, $p = 0.0004$). Most significantly, under heavy congestion (22.2 veh/min), DQN reduces waiting time

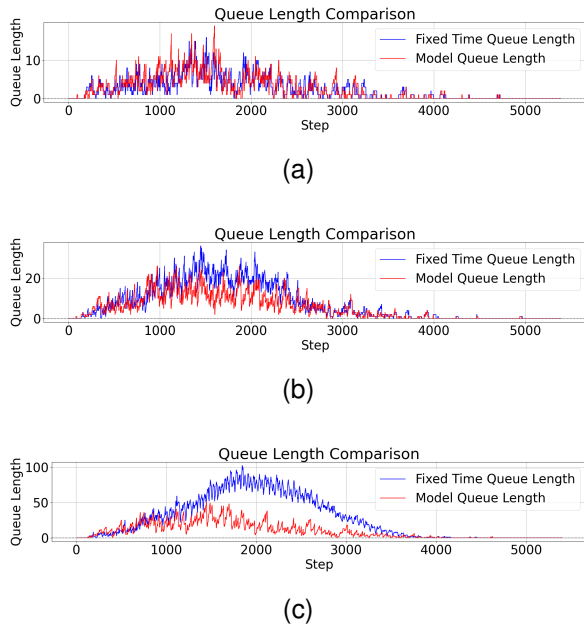


Fig. 5. Per-second queue lengths: fixed-time and DQN controllers (a) 8.3 veh/min traffic scenario, (b) 16.7 veh/min traffic scenario, (c) 22.2 veh/min traffic scenario.

TABLE III
COMPARISON OF TOTAL VEHICLE WAITING TIME BETWEEN DQN AND FIXED-TIME CONTROLLERS.

Traffic Scenario	Controller	Mean Waiting Time (s) $\pm$ std	p -value (Paired t -test)
8.3 veh/min	DQN	12,670 $\pm$ 492	0.041
	Fixed-Time	12,999 $\pm$ 678	
16.7 veh/min	DQN	29,361 $\pm$ 1,124	0.0004
	Fixed-Time	38,310 $\pm$ 6,318	
22.2 veh/min	DQN	57,011 $\pm$ 5,579	7.3×10^{-6}
	Fixed-Time	170,274 $\pm$ 47,565	

by 66.5% (57,011s vs. 170,274s, $p = 7.3 \times 10^{-6}$). Additionally, DQN exhibits substantially lower variance under heavy traffic (std = 5,579 vs. 47,565), indicating more reliable performance. All improvements are statistically significant based on paired t -tests, demonstrating the superior adaptive capability of the DQN controller in high volume traffic conditions.

IV. CONCLUSION AND FUTURE WORK

This study compared a DQN-based adaptive traffic signal controller with a Webster's fixed-time controller under varying traffic demands. The DQN agent demonstrated effective learning and greatly reduced queue lengths and waiting times in 16.7 veh/min and 22.2 veh/min demand scenarios, achieving more than 66% lower delays in heavy traffic. Performance is somewhat improved in the 8.3 veh/min scenario, indicating limited benefits when congestion is minimal. Overall, the results highlight the potential of reinforcement learning-based controllers to improve traffic efficiency, particularly under congested conditions.

However, the proposed system is designed for conventional human-driven traffic, as modeled in the SUMO simulation.

In real-world conditions, driver behavior can be highly unpredictable, such as delayed reactions or non-compliance with signals, which may influence the controller's performance. For practical application, the controller is required to handle practical challenges such as sensor noise, partial observability, and communication delays between detection units and the controller. These factors can introduce uncertainty in the observed state and delay in action execution, potentially degrading control performance. Future work could explore- (i) incorporating multiple vehicle types rather than a single type could better capture heterogeneous traffic dynamics, and (ii) including noise models, delayed state feedback, or communication-aware learning frameworks to improve robustness and ensure reliable operation in real urban environments.

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